

Assignment - 08.

Q1.

→

Steps to create a module in python are as follows -

i) Create a python file (which will be imported).

ii) Create another file to use first file as a module.

iii) In second file use import statement, with name of first/module file.

iv) You can directly use module's functions via `module_name.function_name` operators with module's name / its alias name.

Q2.

→

C, C++ uses `{ }` to define code blocks, whereas Python completely removed concept of curly brackets to define code block, instead they use `:` and indented next lines.

Promotes consistent formatting, Better readability.

Q3.

→

`IndentationError` is raised when Python encounters incorrect indentations.

could be also raised due to mismatch tabs / alignment, unexpected / random indentation.

Q4.

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Ways to import a module are as follows -

i) `import m_name.`

ii) `import m_name as alias_name`

iii) `from m_name import method_name.`

iv) `from m_name import *`

Q5.

- Python's lexer generates two special tokens:
- i) INDENT
 - ii) DEDENT

& interpreter uses these tokens to determine block boundaries. these are figured out during parsing.

Q6.

- "from module-name import *" a bad practice, as during multiple imports, two methods with same name may collide, harder debugging,

Q7.

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- i) A module can be imported into multiple projects
 - ii) Easier testing → Each tiny module
 - iii) Better Scalability
 - iv) Better Collaboration

Q8.

- PVM's internal process after import module:

- i) check if the module already exists in sys.modules
- ii) If YES, → run it the already existing.
- iii) Otherwise - Searches the module in-
 - Current working directory
 - Standard library
 - Installed Packages

- iv Reads & compiles the module.
- v Executes all top level statements.
- vi Creates module's object.
- vii Stores it in `sys.module`
- viii Binds the module name in the current namespace.

Q9.

→ The print statement will execute immediately during import.

```
eg- print ("Hello")
      def add (c):
          pass
```

} demo.py.

import demo → main.py

Output - Hello.

Q10.

→ Modules supports 'scalable softwares like -

- i Proper code set's Organization
- ii Promoting code reuse.
- iii Reducing duplication
- iv Improving readability.
- v Independent testing.
- vi Active easy collaboration.